

**From Coaching to Classroom Effectiveness:A Randomized
Controlled Trial of Instructional Coaching Intensity in the
Teach For Nigeria(TFN) Fellowship**

A Proposal

Nene Onyedi Ibezim

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Abstract

Background: Teach For Nigeria (TFN) is a two-year fellowship program that selects high-potential Nigerian graduates through a competitive process, prepares them through an intensive six-week pre-service training, and places them as full-time teachers in under-resourced public schools. The program's primary ongoing investment in fellow development is individualized instructional coaching delivered throughout both fellowship years. A 2024 evaluation by the RAND Corporation found that students taught by TFN fellows outperformed matched comparison peers by 0.07 standard deviations (SD) in literacy and 0.11 SD in mathematics, and that second-year fellows are more effective than first-year fellows (Mihaly, Schweig, Lin Wang, & Lee, 2024). However, the specific causal contribution of coaching intensity to fellow teaching effectiveness, TFN's most resource-intensive ongoing program component, has never been estimated using a randomized design.

Objective: To estimate the causal effect of high-dose instructional coaching on TFN fellow teaching effectiveness, as measured by a validated classroom observation composite score at twelve months post-placement, relative to fellows receiving the standard coaching protocol.

Design: A two-arm, individual-level randomized controlled trial (RCT). After completing TFN's standard six-week pre-service training and receiving school placements, fellows are randomized 1:1 to high-dose coaching (HDC) or standard-dose coaching (SDC). The individual fellow is the unit of randomization and the primary unit of analysis. The primary endpoint is Month 12.

Sample: Targeting a minimum detectable effect size (MDES) of 0.50 SD at 80% power and $\alpha = 0.05$, and accounting for 20% fellow-level attrition, 160 fellows (80 per arm) will be enrolled at baseline.

Primary Outcome: Composite teaching effectiveness score from a validated classroom observation instrument, assessed at Month 12 by trained, blinded observers.

Significance: This is the first randomized study to estimate the effect of coaching intensity on TFN fellow effectiveness. Results will provide actionable evidence on whether TFN's most substantial ongoing program investment (instructional coaching) produces commensurate improvements in the quality of fellow teaching.

Keywords: Teach For Nigeria; randomized controlled trial; instructional coaching; coaching intensity; teacher effectiveness; sub-Saharan Africa; educational equity; fellowship program

A. Background and Rationale

A.1 Nigeria's Education Crisis

Nigeria confronts one of the most severe learning crises in sub-Saharan Africa. As of 2024, 74 percent of children aged 7 to 14 in Nigeria lack basic reading and mathematics skills, and the country records the highest number of out-of-school primary-age children in the world, with an estimated 10.2 million children not attending school (UNICEF Nigeria, 2024). The structural root of this crisis is well documented: the Universal Basic Education Commission (UBEC) 2022/2023 National Personnel Audit found a deficit of 194,876 primary school teachers in public primary schools nationwide, out of a required total of 694,078, leaving only 499,202 in post, a gap of approximately 28 percent (UBEC, 2024, as reported in Business Day, 2024). The UNESCO Teacher Shortage Report 2023 placed Nigeria's teacher-pupil ratio at 1:35, far above the UNESCO standard of 1:25, with some rural and conflict-affected schools managing ratios approaching 1:124 (UNESCO, 2023, as cited in Premium Times, 2025).

These deficits are not uniformly distributed. Eighteen Nigerian states failed to recruit a single teacher across the five-year period from 2019 to 2024, and the Teachers Registration Council of Nigeria has repeatedly flagged that the shortage is most pronounced in rural communities, precisely those communities that TFN targets (Punch Newspapers, 2024; Athena Centre for Policy and Leadership, 2025). Teacher quality, rather than physical infrastructure or textbooks, is the most proximate determinant of student learning outcomes in developing-country contexts (Glewwe & Muralidharan, 2016). Improving both the supply and effectiveness of teachers in Nigeria's under-resourced schools is therefore the highest-leverage intervention available.

A.2 The Teach For Nigeria Fellowship

Teach For Nigeria (TFN) was founded in 2017 as Nigeria's member of the global Teach For All network, which as of 2024 comprises more than 60 independently governed national organizations working to develop teacher-leaders in underserved communities across the world (Teach For All, 2024). The TFN fellowship is a two-year, full-time, paid commitment in which competitively selected Nigerian graduates and professionals are placed as teachers in public or low-income private schools in Lagos and Ogun States (Teach For Nigeria, 2025a). As of 2025, TFN has received over 79,000 applications, placed 2,965 fellows across more than 128 higher education institutions of origin, and reached an average of 140 students per fellow (Teach For Nigeria, 2025b).

The fellowship has three structural components after initial selection. First, a six-week intensive pre-service training covers 21st-century teaching pedagogy, leadership, classroom management, and entrepreneurship, preparing fellows for placement before the school year begins (Teach For Nigeria, 2025a). Second, fellows are placed in schools in Ogun or Lagos State, where they serve as full-time teachers either as class teachers across all subjects or as subject-matter specialists in English, mathematics, or science (Teach For Nigeria, 2025b). Third, TFN provides ongoing instructional coaching throughout both fellowship years, including classroom observations, performance reviews, and data-informed feedback from TFN coaches. It is this third component (coaching) that constitutes TFN's primary continuous investment in fellow development and that is the subject of this trial.

TFN's fellowship is explicitly designed as a leadership development program, not only a teacher placement initiative. Fellows commit to two years of full-time teaching while simultaneously

developing leadership skills, stakeholder management, planning, execution, and critical reflection that are intended to equip them for long-term contributions to education equity beyond the classroom (Teach For Nigeria, 2025a). After the fellowship, TFN alumni enter a structured alumni movement currently comprising more than 835 members who continue to work toward systemic change in Nigeria's education sector (Teach For Nigeria, 2025a).

A.3 Evidence on TFN Fellowship Effectiveness

The most methodologically rigorous external evaluation of TFN to date is a two-year mixed-methods study conducted by the RAND Corporation between the 2021–2022 and 2022–2023 school years. The RAND team collected data from 80 head teachers, 183 teachers, and approximately 5,900 students across 80 schools in Ogun State, 40 treatment schools with TFN teachers and 40 comparison schools matched on student demographic characteristics and geographic proximity (Mihaly et al., 2024). Their key quantitative finding was that students taught by TFN fellows performed better in both literacy and mathematics than students in comparison classrooms: the literacy advantage was approximately 0.07 SD and the mathematics advantage was approximately 0.11 SD, with both results statistically significant at $p < 0.05$ (Mihaly et al., 2024). Qualitative findings reinforced these estimates: parents, head teachers, and non-TFN teachers all described substantive improvements in students' reading, writing, English-language fluency, and mathematical reasoning that they attributed to TFN fellows (Mihaly et al., 2024).

The RAND evaluation also documented a year-in-program effect: second-year TFN teachers were more effective than their first-year peers, a difference that was statistically significant for mathematics achievement (Mihaly et al., 2024). The RAND team interpreted this as evidence

that accumulated classroom experience within the fellowship supported by ongoing coaching and professional development drives effectiveness growth over time. However, the RAND design could not identify which specific component of the fellowship drives this growth. The evaluation used a comparison-group rather than a randomized design, and it was not structured to isolate the causal contribution of coaching visits, coaching quality, coaching frequency, or any other specific program element.

Beyond TFN, evidence on the Teach For All model more broadly comes primarily from quasi-experimental evaluations. In the United States, the most rigorous evidence on Teach For America (TFA), the Teach For All network's founding member, comes from a randomized evaluation by Decker, Mayer, and Glazerman (2004) in which students were randomly assigned to TFA or non-TFA teachers within the same school. That study found TFA teachers produced gains of approximately 0.15 SD in mathematics relative to non-TFA comparison teachers, with no significant reading effect (Glazerman, Mayer, & Decker, 2006). Subsequent randomized evaluations of TFA have found similarly positive mathematics effects and mixed reading effects (Clark et al., 2013). A 2025 RAND evaluation of Teach For Pakistan found comparable patterns of positive effects on student academic outcomes combined with inconclusive social-emotional learning effects, suggesting some generalizability across Teach For All contexts (Schweig, Lin Wang, Lee, & Mihaly, 2025). Nevertheless, none of these evaluations in Nigeria, the United States, or elsewhere has randomized coaching intensity as a treatment condition. The question of whether more coaching produces better fellows remains unanswered.

A.4 Instructional Coaching as a Program Component

Instructional coaching in which a trained coach provides individualized, classroom-observation-based feedback to a teacher on a recurring basis has been identified as one of the most promising professional development modalities in the educational research literature. Kraft, Blazar, and Hogan (2018) conducted a meta-analysis of 60 studies employing causal research designs and found pooled effect sizes of 0.49 SD on instructional practice and 0.18 SD on student achievement. These are among the largest documented effects of any teacher development intervention. Critically, the Kraft et al. (2018) meta-analysis also found that effects at scale in larger programs tended to be smaller than effects in smaller efficacy trials, and that the quality and specificity of coaching, not simply volume, may be a moderating factor.

However, a summary in Learning Professional noted that within the Kraft et al. (2018) meta-analysis, there was no clear evidence that more hours of coaching were associated with larger impacts, suggesting quality and focus may be more important than quantity alone (ERIC, 2018). This finding complicates simple predictions about the effects of increasing coaching dose. Whether doubling coaching frequency within a specific program context such as TFN's Ogun State operations produces proportionate gains in fellow effectiveness is a genuinely open empirical question, and one that cannot be answered without exogenous variation in coaching dose. This trial is designed to provide that variation through randomization.

A.5 Rationale for Randomization

Observational analyses of coaching effects within TFN's program would face intractable confounding. Fellows who receive more coaching may be those who are struggling and require remediation, or conversely those who are geographically accessible and therefore easier to reach.

Either pattern would create a correlation between coaching dose and fellow effectiveness that does not reflect a causal relationship. An RCT, in which coaching dose is randomly assigned to fellows who are otherwise identical in their pre-service preparation and school placement, provides the only credible basis for isolating the coaching effect. Because TFN places exactly one fellow per school, individual randomization of fellows is feasible without any within-school contamination: a fellow in the high-dose condition and a fellow in the standard-dose condition are always in entirely separate schools with no shared instructional environment.

B. Study Objectives and Research Questions

B.1 Primary Objective

To estimate the causal intent-to-treat (ITT) effect of assignment to high-dose instructional coaching on TFN fellow teaching effectiveness, measured by a validated classroom observation composite score at twelve months post-placement, relative to assignment to standard-dose coaching.

Primary Research Question: *Does random assignment to high-dose coaching (HDC) produce a significantly higher teaching effectiveness score at Month 12 than random assignment to standard-dose coaching (SDC)?*

B.2 Secondary Objectives

- **Self-reported pedagogical effectiveness:** Does assignment to HDC improve fellows' self-reported confidence in lesson planning, use of formative assessment, and responsiveness to student needs, assessed at Month 6 and Month 12?

- **Effectiveness growth trajectory:** Does the gain in observed teaching effectiveness between Month 6 and Month 12 differ significantly between HDC and SDC fellows, suggesting that higher coaching dose accelerates within-year instructional development?
- **Manipulation check:** Did HDC fellows receive meaningfully more coaching visits than SDC fellows, confirming that the randomized contrast in coaching dose was successfully implemented?

C. Conceptual Framework and Theory of Change

C.1 The Teach For All Theory of Change

TFN is a member of the Teach For All global network, whose theory of change rests on the premise that individuals with demonstrated leadership potential, selected through a rigorous competitive process, can, if prepared and supported adequately, become highly effective teachers in under-resourced schools and thereafter long-term advocates for educational equity (Teach For All, 2024). Approximately 50 to 70 percent of Teach For All alumni worldwide remain in education-related careers long-term, consistent with the network's goal of building a sustained leadership movement (Teach For All, 2024). Within this overarching theory, the fellowship is designed to work through three temporal stages.

In the short term, roughly the first fellowship year, the mechanism is primarily one of intensive preparation and supported practice: fellows enter classrooms with stronger foundational pedagogical knowledge than traditionally trained teachers, and ongoing coaching from TFN coaches is designed to accelerate early instructional skill development. In the medium term, the second fellowship year and immediately beyond the theory anticipates that accumulated

experience and coaching produce a step-change in effectiveness, consistent with the year-in-program finding of the RAND evaluation (Mihaly et al., 2024). In the long term, alumni leverage their classroom experience and leadership development to drive systemic change in the education sector (Teach For Nigeria, 2025a).

C.2 The Causal Pathway Under Test

This trial tests the following specific causal pathway: random assignment to HDC → more frequent structured observation and individualized feedback → accelerated improvement in instructional skill between observation cycles → higher observed teaching effectiveness at Month 12 relative to SDC peers. Every element of the fellowship other than coaching frequency is held constant across both arms. Fellows in both conditions are selected by the same process, trained by the same six-week curriculum, and placed by the same school-matching procedures. Coaching intensity is the only variable that differs. The trial therefore isolates coaching dose as the causal agent.

C.3 What the Trial Does Not Test

This trial does not test whether TFN fellows are more effective than non-TFN teachers in the same schools, that question has been addressed by the RAND evaluation using a matched comparison design (Mihaly et al., 2024). It does not test whether pre-service training quality affects fellow effectiveness, nor whether school-level administrator support moderates TFN outcomes. These are important questions for future research but are not addressable within the current design, which requires all pre-placement components to be held constant across randomized arms.

D. Study Setting

The study will be conducted in public and low-income private schools across Ogun State, Nigeria, where TFN is actively placing fellows. Ogun State was the exclusive site of the RAND evaluation (Mihaly et al., 2024), and it is one of TFN's two primary operating locations alongside Lagos State. The state encompasses both peri-urban areas adjacent to Lagos and more rural communities, creating a range of school contexts within a single administratively coherent geography.

Schools eligible for TFN placement in Ogun State are, by TFN's own programme criteria, under-resourced and teacher-scarce representing precisely the communities documented in the UBEC (2024) teacher shortage data. TFN maintains a working partnership with Ogun State government for school selection and fellow placement (Teach For Nigeria, 2025b). This existing infrastructure including TFN's coaching visit scheduling systems, administrative data collection on fellow performance, and relationships with school head teachers means that the trial can be embedded within TFN's standard operating processes without requiring the establishment of new institutional relationships or data collection systems.

All study activities will take place during the regular school year, coordinated with TFN's annual program calendar, which begins with the six-week pre-service training before the academic year opens and continues with regular coaching cycles through the year (Teach For Nigeria, 2025a).

E. Target Population and Context

E.1 Fellows

The trial population is TFN fellows recruited for a given annual cohort and placed in Ogun State schools. A fellow is eligible for enrollment in the trial if he or she: (a) has completed all stages of TFN's competitive selection process and formally accepted a fellowship offer; (b) has completed the six-week pre-service training; (c) has received a school placement in Ogun State for the upcoming fellowship year; and (d) provides written informed consent for inclusion in trial data collection prior to randomization. There are no further exclusion criteria.

[Assumption: A single annual TFN cohort in Ogun State produces between 80 and 120 placed fellows. If a single cohort does not produce 160 consenting enrollees, the trial will continue into the following annual cohort. Cohort year will be recorded as a stratification and covariate variable in this case. This scale is consistent with TFN's documented reach of 2,965 total fellows across all cohorts and both states, and with TFN's stated goal of reaching 20,000 leaders over the next decade (Teach For Nigeria, 2025a).]

E.2 Description of the Intervention Condition: High-Dose Coaching (Treatment Arm)

Fellows randomized to the HDC arm receive instructional coaching at twice the frequency of TFN's current standard coaching protocol throughout Fellow Year 1. If TFN's current standard is an average of one structured coaching visit per month per fellow across the school term, HDC fellows receive an average of two visits per month. Each visit follows TFN's established coaching protocol: a structured classroom observation using a standardized observation guide, a post-observation debrief with the fellow, a review of student progress data, and collaborative goal-setting for the period until the next observation (Teach For Nigeria, 2025a).

The content structure and quality standards of individual coaching sessions are identical across both arms. Only frequency differs. This design choice is deliberate: by holding session content constant and varying only dose, the trial's estimate of the HDC effect is attributable specifically to the additional quantity of coaching interactions, not to any qualitative difference in coaching approach. In this way, the trial tests the marginal return to coaching volume, a direct input into TFN's cost-effectiveness calculus.

[Assumption: TFN's coaching staff in Ogun State can sustain twice the standard visit frequency for the HDC arm, which constitutes approximately half the total trial enrollment, without degrading individual session quality. This is an operational assumption that will be confirmed with TFN coaching leadership before trial launch. Pre-trial coach capacity assessment and a rolling fidelity monitoring plan will be used to detect and respond to any coaching quality deterioration.]

E.3 Description of the Comparison Condition: Standard-Dose Coaching (Comparison Arm)

Fellows randomized to the SDC arm receive TFN's currently operational, standard-frequency coaching protocol, the programme as it already exists. The SDC arm is not a no-coaching condition. Fellows receive coaching at TFN's current standard rate, with the same session structure, observation protocol, and performance review procedures as the HDC arm. The difference is solely the number of sessions delivered per term.

E.4 What Both Arms Receive Equally

Both HDC and SDC fellows receive the complete standard TFN fellowship: the competitive selection process, the six-week pre-service training, school placement support, leadership and community engagement programming, access to the TFN alumni movement, and the stipend provided throughout the fellowship year (Teach For Nigeria, 2025a). No fellow in either arm is denied anything that is part of TFN's standard programme offer.

F. Sample Size and Power Analysis

F.1 Study-Powering Strategy

The study is powered on the primary outcome of observed teaching effectiveness at Month 12, assessed at the individual fellow level. Because the unit of randomization is the individual fellow and TFN places one fellow per school, there is no clustering of fellows within randomized units. The power calculation for this trial is therefore an individual-level, two-arm comparison requiring no design effect correction. This is the central statistical simplification of a fellow-focused design relative to a school-level cluster-randomized trial.

F.2 Effect Size Target

The meta-analytic benchmark for individualized instructional coaching effects on teaching practice is a pooled effect size of 0.49 SD across 60 causally identified studies (Kraft, Blazar, & Hogan, 2018). This estimate encompasses a range of coaching contexts and program types. The proposed intervention, doubling coaching frequency within an ongoing fellowship program that already provides some coaching is unlikely to produce effects as large as those from a full coaching-versus-no-coaching comparison. The study is powered to detect an effect of 0.50 SD,

which is approximately equal to the meta-analytic benchmark of 0.49 SD for coaching effects on instructional practice. This is a conservative target because the intervention compares high-dose coaching to standard-dose coaching, rather than coaching to no coaching

This target is also calibrated against Kraft's (2020) empirical benchmarks for education interventions, which classify an effect of 0.20 SD or greater as large in the field-based education research context. An MDES of 0.50 SD is therefore conservative relative to what would be needed for statistical significance but represents a threshold at which TFN could justify a major investment in coaching expansion. The study is not powered to detect effects below 0.50 SD on the teaching observation composite; effects in the 0.20–0.40 SD range, if observed, would be informative but would require replication with a larger sample.

F.3 Required Sample Size

For a two-arm individual-level RCT with equal group sizes, two-sided alpha = 0.05, and 80% power, the required per-arm sample size is:

$$n \text{ per arm} = 2 \times (z_{\{\alpha/2\}} + z_{\{\beta\}})^2 / d^2$$

Substituting $z_{\{\alpha/2\}} = 1.96$, $z_{\{\beta\}} = 0.842$, $d = 0.50$:

$$n \text{ per arm} = 2 \times (2.802)^2 / 0.25 = 2 \times 7.851 / 0.25 \approx 63 \text{ fellows per arm}$$

Rounding to 64 per arm for symmetry, the pre-attrition analytic target is 128 fellows. Accounting for a pre-specified 20% fellow-level attrition rate, reflecting fellows who may withdraw from the fellowship mid-year, be dismissed, or be otherwise unavailable for the Month 12 observation, the enrollment target is: Enrollment target = $128 / 0.80 = 160$ fellows total (80 per arm)

F.4 Power Parameter Summary

Parameter	Value and Justification
Significance level (α)	0.05, two-tailed
Power ($1 - \beta$)	0.80
MDES	0.50 SD (conservative relative to Kraft et al., 2018 meta-analytic mean of 0.49 SD for coaching effects on practice)
Design	Individual-level RCT; no clustering correction required
Fellows per arm (pre-attrition)	64
Assumed fellow attrition rate	20%
Fellows per arm to enroll	80
Total fellows to enroll	160

G. Data Collection and Measurement

G.1 Data Collection Timeline

Time Point	Target	Instrument / Source	Method
Baseline (Month 0)	Fellows	Demographic and background survey; prior teaching experience; school type; cohort year	Self-report survey
Month 3	TFN Admin	Coaching visit logs (fidelity check, blinded to outcomes)	Administrative records

Month 6	Fellows	Classroom observation (scheduled + unannounced); fellow self-report survey (effectiveness and coaching satisfaction items)	Blinded observation; online survey
Month 12	Fellows	Classroom observation (scheduled + unannounced); fellow self-report survey; coaching visit log (full Year 1)	Blinded observation; survey; admin records

G.2 Classroom Observation

Trained observers will conduct two classroom observation visits per fellow per assessment wave (Month 6 and Month 12), one scheduled with 48-hour advance notice, and one unannounced. Observers will complete the standardized observation instrument for each visit, and the composite teaching effectiveness score for that wave will be the mean of the two-visit item scores. Observers will receive two full days of standardized training and a calibration session before each data collection wave. A 10% random subsample of sessions will be double rated by a second observer for inter-rater reliability monitoring. If inter-rater agreement on the composite falls below 0.75 at any point during data collection, collection will pause and recalibration will be conducted before resuming. Observers will have no access to any TFN records about fellows' coaching conditions, visit counts, or performance reviews.

G.3 Fellow Self-Report Survey

A structured online survey will be administered to fellows at Month 6 and Month 12. The survey will include items covering: self-rated confidence in lesson planning and delivery; self-rated use of student performance data to adjust instruction; self-assessed improvement in teaching skills

over the fellowship period; and perceived quality and actionability of coaching feedback. The coaching satisfaction items serve as a manipulation check, confirming that HDC fellows reported perceiving more coaching engagement than SDC fellows. Paper versions will be available for fellows with limited internet connectivity. Survey completion will be monitored by the research team, with one reminder sent to non-responders at seven days after initial administration.

G.4 Administrative Data on Coaching Visits

TFN's program administration team will provide, for each participating fellow on a termly basis: the number of coaching visits received, the number of formal performance review sessions conducted, and fellow placement and retention status. These data will be used as the manipulation check variable, confirming that the randomized contrast in coaching dose was successfully implemented and as implementation fidelity indicators for the sensitivity analyses described in Section O.

H. Statistical Analysis Plan

H.1 Primary Analysis: Intent-to-Treat

The primary analysis estimates the intent-to-treat (ITT) effect of coaching arm assignment on the Month 12 observed teaching effectiveness composite. ITT is defined as the effect of assignment to HDC (regardless of the number of coaching visits received) versus assignment to SDC. The ITT estimand is preferred as the primary target because it reflects the causal effect of the policy of assigning fellows to higher coaching frequency, the question most relevant to TFN's program design decisions rather than the effect conditional on perfect protocol adherence.

The estimator is ordinary least squares (OLS) regression of the Month 12 composite score on a binary indicator for coaching arm (1 = HDC, 0 = SDC), controlling for the following pre-specified baseline covariates: stratification variables (cohort year, school type, prior teaching experience), and the Month 6 teaching effectiveness composite where available. Pre-specifying baseline covariates increases precision by absorbing residual variance and is standard practice in RCT analysis.

H.2 Manipulation Check

Before reporting primary results, the analysis will confirm that the randomized coaching frequency contrast was implemented as designed. The mean number of coaching visits received per term will be compared between HDC and SDC fellows using a two-sample t-test on the administrative visit log data.

H.3 Missing Data and Attrition

Fellows who leave the fellowship before Month 12 will not contribute a primary outcome observation. Their baseline characteristics will be compared between arms to assess whether attrition is differential.

I. Moderation Analysis Plan

Two pre-registered exploratory moderation analyses will test whether the effect of coaching dose differs across pre-specified subgroups. Both are designated exploratory given the insufficient power for interaction testing at the proposed sample size. Results will be presented with effect size estimates and 90% confidence intervals rather than binary significance tests and will be clearly labeled as hypothesis-generating.

- **School type as moderator:** Does the effect of HDC on teaching effectiveness differ between fellows placed in primary schools versus secondary schools? This is motivated by the observation that TFN places fellows across both school levels (Teach For Nigeria, 2025b), and that coaching needs and observation dynamics may differ substantially between primary and secondary classroom settings. A significant interaction would suggest that HDC dose effects are context-dependent and that coaching investment should be differentially targeted by school level.
- **Prior teaching experience as moderator:** Do fellows with no prior teaching experience benefit more from HDC than fellows who enter with some classroom experience? If coaching is most valuable for those who lack foundational exposure to classroom management and instructional pacing, TFN could target intensive coaching toward inexperienced fellows and reduce visit frequency for more experienced ones, achieving efficiency gains without sacrificing effectiveness. The RAND evaluation noted that most TFN fellows do not have academic backgrounds in education or prior teaching experience (Mihaly et al., 2024), but within-cohort variation in prior experience provides the within-trial variation needed for this test.

No mediation analysis is pre-specified. The self-report coaching satisfaction items (administered at Month 6 and Month 12) will be used descriptively to explore whether fellows' perceived quality of coaching differs between arms, consistent with the hypothesis that more coaching visits produce a stronger coach-fellow developmental relationship.

J. Ethical Considerations

J.1 Ethical Approval

The trial will require ethical review and approval from the Institutional Review Board (IRB) or Research Ethics Committee of the principal investigator's home institution, and from the National Health Research Ethics Committee of Nigeria (NHREC). No data collection will begin before all required approvals are obtained and documented. The trial will be registered on the AEA RCT Registry prior to randomization.

J.2 Informed Consent

All participating fellows will provide written informed consent before enrollment. Consent forms will clearly explain the study purpose, the randomization procedure, the existence of two coaching conditions, what data will be collected and how it will be stored, and the fellow's unconditional right to withdraw from the study at any time without any consequence for their fellowship standing, placement, or stipend. Research team members, not TFN coaching or program management staff will conduct all consent processes to prevent any real or perceived coercion arising from TFN staff involvement.

J.3 Ethical Justification of the SDC Comparator

The comparison condition in this trial is TFN's existing, operational standard coaching protocol. Every fellow in the study receives coaching at no less than the rate they would have received in the absence of the trial. The SDC arm does not withhold any programme component that fellows are otherwise entitled to. The ethical concern common to trials in which a control group receives no treatment, that withholding a beneficial intervention causes harm does not arise here. Genuine scientific uncertainty about whether doubling coaching frequency produces meaningfully better fellows, the question this trial is designed to answer is a fully sufficient condition for the ethical permissibility of the comparison.

J.4 Data Privacy

All fellow-level data will be collected and stored with a unique de-identified participant identifier. Personally identifiable information, names, school identification, and contact details will be stored separately from outcome data in a password-protected, encrypted repository accessible only to the principal investigator and study statistician. Fellow survey responses will be kept strictly confidential from TFN coaching staff, program management, and any TFN evaluators. This confidentiality assurance is essential for the validity of self-report data. Fellows must feel safe to respond honestly about their perceptions of coaching quality without fear of professional consequences.

K. Expected Contributions

This trial makes three distinct contributions to the literature on fellowship-based teacher development programs.

First, it provides TFN with the first randomized evidence on the causal effect of coaching intensity on fellow teaching effectiveness. The RAND evaluation (Mihaly et al., 2024) established that TFN fellows produce positive student learning outcomes on average; this trial identifies whether the program's most resource-intensive ongoing component (coaching) is a driver of those outcomes and whether increasing its dose would improve returns. A positive finding would justify coaching staff expansion; a null finding would redirect TFN's investment toward other program levers such as pre-service training depth or school administrator preparation.

Second, the study contributes to the international literature on instructional coaching in low- and middle-income country contexts. The Kraft, Blazar, and Hogan (2018) meta-analysis drew predominantly on evidence from the United States and other high-income settings. The dose-response question, whether more coaching visits produce proportionately better teaching, has not been studied in a randomized design in sub-Saharan Africa. A well-powered RCT in Ogun State would provide the first causal evidence on this question from the region.

Third, the design itself, an individual-level RCT embedded within an ongoing fellowship program, using coaching dose as the randomized treatment demonstrates a feasible approach to component-testing in complex teacher development programs. It can serve as a methodological template for future within-program evaluations in the Teach For All network and beyond.

L. Generalizability and Limitations

L.1 Generalizability

The study's findings will be most directly applicable to TFN's Ogun State operations. Given the broadly similar socioeconomic and educational context between Ogun and Lagos States (TFN's other primary operating location) cautious generalization to Lagos State is reasonable. Generalization to Teach For All network organizations in other LMICs is plausible to the extent that the coaching structure, fellow profile, and school placement context are comparable; Teach For Pakistan, Teach For India, and Teach For Uganda share the core fellowship architecture. Generalization to Teach For America or other high-income-country network members should not be assumed, given the substantially different teacher labor market, school resource environment, and coaching infrastructure in those contexts.

L.2 Limitations

MDES of 0.50 SD is larger than the documented TFN student outcome effects: The RAND evaluation found TFN fellows produce 0.07–0.11 SD gains in student learning relative to comparison students (Mihaly et al., 2024). Effects of HDC coaching on student outcomes, two steps removed from the randomized treatment are likely to be much smaller than effects on fellow teaching practice. The exploratory student literacy analysis at this sample size will be uninformative for anything smaller than approximately 0.40 SD, which is implausibly large for a coaching dose effect on student outcomes.

Fellow Year 1 only: The primary endpoint is at twelve months, covering only the first fellowship year. The RAND evaluation found that second-year fellows are substantially more effective than first-year fellows (Mihaly et al., 2024), suggesting that coaching effects may compound across the full two years. A twelve-month window may underestimate the cumulative value of HDC. Year 2 follow-up is a priority for subsequent research.

Consent-based enrollment: Fellows who decline to consent may systematically differ from those who enroll. If non-consenting fellows are those least confident in their teaching skills, or most concerned about observer judgment, the consenting sample may be positively selected. Baseline characteristics of consenting and non-consenting fellows will be compared and reported to the extent that TFN's administrative data permit.

Single-program context: This trial estimates the effect of HDC within a specific fellowship program with a specific pre-service training curriculum, specific coaching staff, and specific school placement context. The estimated effect is specific to TFN's Ogun State program and

should not be treated as a general coefficient on coaching frequency that transfers across all teacher development programs.

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